

HARSHAL GUPTA

91-9151448889 Uttar Pradesh, India

harshal.2023ug3013@iiitranchi.ac.in [LinkedIn](#) [Medium](#) [GitHub](#) [Codeforces](#) [LeetCode](#) [CodeChef](#)

EDUCATION

B.Tech in Computer Science with Specialization in Data Science and AI
Indian Institute of Information Technology, Ranchi

CGPA: 8.07
2023–2027

SKILLS

- **Languages:** Java, C, Python, TypeScript, JavaScript
- **Web & Tools:** React, Node.js, Flask, Express.js, Tailwind CSS, FastAPI and Pydantic, Git, Postman, VS Code, GitHub Copilot, SQL, MongoDB.
- **Other:** JWT Auth, OAuth, Clerk ,REST APIs, Linux, Networking, Numpy, Pandas, Matplotlib, AWS, GCP.

EXPERIENCE

dAaranya- Full Stack Developer Intern

Sep 25 - Mar 26

- Architected a scalable **FastAPI** backend featuring rigorous **Pydantic** data validation and a core **Pub-Sub model**, driving key technical decisions and deploying via **CI/CD pipelines on GCP**.
- Optimized **PostgreSQL** database throughput for **100+ users** via **connection pooling**, integrated creational design patterns (**Builder, Singleton, Factory**), and **slashed LLM API costs by 50%** using the **Batch API**.

PROJECTS

Medic - Full Stack Medical Management System

[\(GitHub\)](#)

- Designed a scalable **RESTful backend** using **Flask** and **MongoDB**, reducing development time by **35%** through modular design for sustaining and managing healthcare.
- Built an **AI-based health chatbot** with **85% accuracy** using trained models and integrated real-time backend support with vector embeddings and **CassandraDB**.

Language Learning App -Learn Japanese (Ongoing)

[\(GitHub\)](#)

- Created a **MERN+Typescript Gamified App** using **TypeScript, React, Prisma, and Redis**.
- Implemented user authentication, **OpenAI TTS and STT**, used **Spaced Repetition Algorithms** to boost learning retention.
- Designed scalable **RESTful backend and websockets** and enhancing user engagement.

HONORS AND ACHIEVEMENTS

- **2nd Place**, InternLay Hackathon among **20+ participants** at IIIT Ranchi.
- **2nd Runner Up**, BITS Pilani Postman Hackathon.
- **Best Backend Developer** for project using **MERN, Flask, and vector DB**, chosen from **20+ teams**.
- **College Rank 4** in DP and Graphs contest by AlgoUniversity.
- **GSSoC 2025 Contributor**, contributed to open source projects in GirlScript Summer of Code 2025.
- Solved **1300+ questions** across **LeetCode, GFG, Codeforces, and CodeChef**.
- **Knight on LeetCode**, **3-Star on CodeChef**, and **Specialist on Codeforces**.
- **Partnership Co-Lead GDG** - Led partnership initiatives and community engagement.